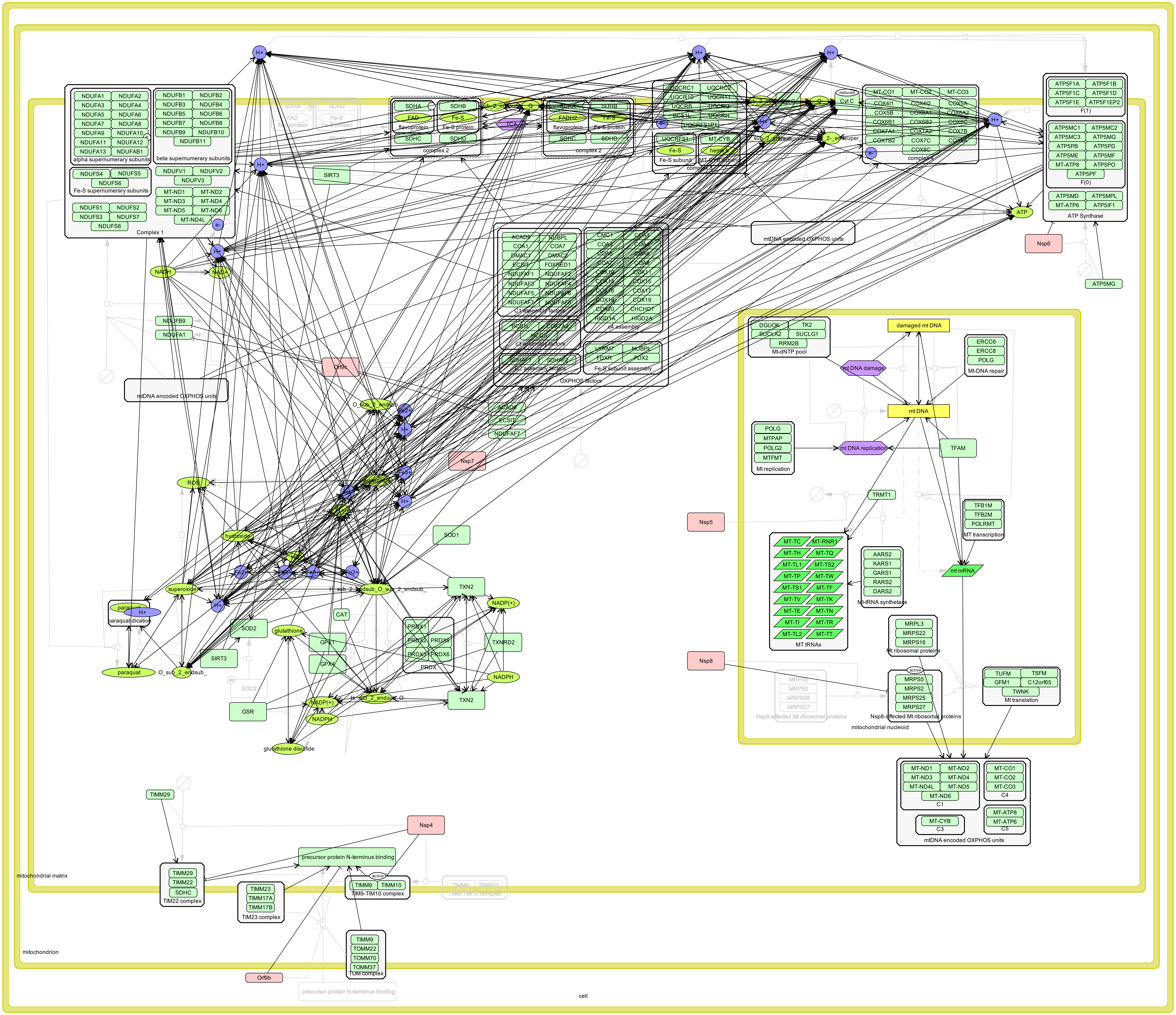
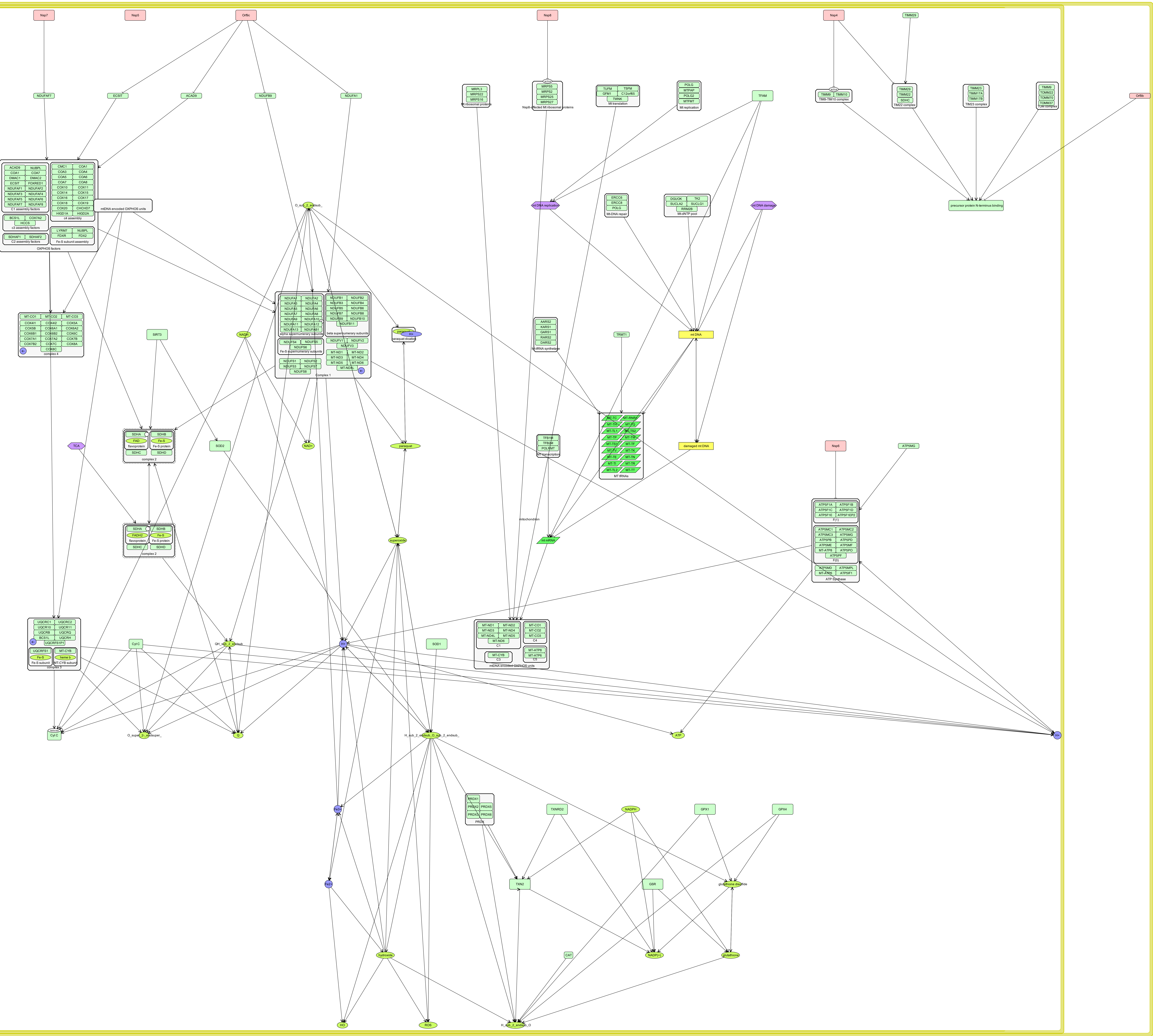

H+
H+
H+
NDUFA1
NDUFA2
NDUFA3
NDUFA4
NDUFA5
NDUFA6
NDUFA7
NDUFA8
NDUFA9
NDUFA10
NDUFA11
NDUFA12
NDUFA13
NDUFAB1
alpha supernumerary subunits
NDUFB1
NDUFB2
NDUFB3
NDUFB4
NDUFB5
NDUFB6
NDUFB7
NDUFB8
NDUFB9
NDUFB10
NDUFB11
beta supernumerary subunits
NDUFS4
NDUFS5
NDUFS6
Fe-S supernumerary subunits
NDUFV1
NDUFV2
NDUFV3
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND5
MT-ND6
MT-ND4L
NDUFS1
NDUFS2
NDUFS3
NDUFS7
NDUFS8
e-
Complex 1
SDHA
SDHB
FAD
Fe-S
flavoprotein
Fe-S protein
SDHC
SDHD
complex 2
O_sub_2_endsub
Q
TCA
SDHA
SDHB
FADH2
Fe-S
flavoprotein
Fe-S protein
SDHC
SDHD
complex 2
UQCRC1
UQCRC2
UQCR10
UQCR11
UQCRB
UQCRQ
BCS1L
UQCRH
UQCRFS1P1
e-
UQCRFS1
Fe-S
Fe-S subunit
MT-CYB
heme b
MT-CYB subunit
complex 3
H+
O_sub_2_endsub
O_sub_2-_endsuper
reduced
Cyt C
Q
MT-CO1
MT-CO2
MT-CO3
COX4I1
COX4I2
COX5A
COX5B
COX6A1
COX6A2
COX6B1
COX6B2
COX6C
COX7A1
COX7A2
COX7B
COX7B2
COX7C
COX8A
COX8C
e-
complex 4
H+
ATP5F1A
ATP5F1B
ATP5F1C
ATP5F1D
ATP5F1E
ATP5F1EP2
F(1)
ATP5MC1
ATP5MC2
ATP5MC3
ATP5MG
ATP5PB
ATP5PD
ATP5ME
ATP5MF
MT-ATP8
ATP5PO
ATP5PF
F(0)
ATP5MD
ATP5MPL
MT-ATP6
ATP5IF1
ATP Synthase
ATP
Nsp6
ATP5MG
SIRT3
H+
NADH
NAD+
NDUFB9
NDUFA1
mtDNA encoded OXPHOS units
Orf9c
ACAD9
NUBPL
COA1
COA7
DMAC1
DMAC2
ECSIT
FOXRED1
NDUFAF1
NDUFAF2
NDUFAF3
NDUFAF4
NDUFAF5
NDUFAF6
NDUFAF7
NDUFAF8
C1 assembly factors
BCS1L
COX7A2
HCCS
c3 assembly factors
SDHAF1
SDHAF2
C2 assembly factors
CMC1
COA1
COA3
COA4
COA5
COA6
COA7
COA8
COX10
COX11
COX14
COX15
COX16
COX17
COX18
COX19
COX20
CHCHD7
HIGD1A
HIGD2A
c4 assembly
LYRM7
NUBPL
FDXR
FDX2
Fe-S subunit assembly
OXPHOS factors
mtDNA encoded OXPHOS units
ACAD9
ECSIT
NDUFAF7
Nsp7
O_sub_2_endsub
Fe2+
H+
Fe3+
superoxide
H+
ROS
H_sub_2_endsub_O
hydroxide
HO
Fe2+
Fe3+
Fe3+
Fe2+
SOD1
superoxide
H+
H_sub_2_endsub_O_sub_2_endsub_
TXN2
NADP(+)
paraquat
H+
paraquat dication
paraquat
O_sub_2_endsub_
SOD2
SIRT3
SOD2
CAT
glutathione
GPX1
GPX4
PRDX1
PRDX2
PRDX5
PRDX3
PRDX6
PRDX
TXNRD2
NADPH
GSR
NADP(+)
NADPH
H_sub_2_endsub_O
TXN2
glutathione disulfide
DGUOK
TK2
SUCLA2
SUCLG1
RRM2B
Mt-dNTP pool
damaged mt DNA
mt DNA damage
ERCC6
ERCC8
POLG
Mt-DNA repair
mt DNA
POLG
MTPAP
POLG2
MTFMT
Mt replication
mt DNA replication
TFAM
Nsp5
TRMT1
TFB1M
TFB2M
POLRMT
MT transcription
MT-TC
MT-RNR1
MT-TH
MT-TQ
MT-TL1
MT-TS2
MT-TP
MT-TW
MT-TS1
MT-TF
MT-TV
MT-TK
MT-TE
MT-TN
MT-TI
MT-TR
MT-TL2
MT-TT
MT tRNAs
AARS2
KARS1
GARS1
RARS2
DARS2
Mt-tRNA synthetase
mt mRNA
MRPL3
MRPS22
MRPS16
Mt ribosomal proteins
Nsp8
MRPS5
MRPS2
MRPS25
MRPS27
Nsp8-affected Mt ribosomal proteins
active
MRPS5
MRPS2
MRPS25
MRPS27
Nsp8-affected Mt ribosomal proteins
TUFM
TSFM
GFM1
C12orf65
TWNK
Mt translation
mitochondrial nucleoid
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
C1
MT-CO1
MT-CO2
MT-CO3
C4
MT-CYB
C3
MT-ATP8
MT-ATP6
C5
mtDNA encoded OXPHOS units
TIMM29
Nsp4
precursor protein N-terminus binding
TIMM29
TIMM22
SDHC
TIM22 complex
TIMM23
TIMM17A
TIMM17B
TIM23 complex
active
TIMM9
TIMM10
TIM9-TIM10 complex
TIMM9
TIMM10
TIM9-TIM10 complex
mitochondrial matrix
TIMM9
TOMM22
TOMM70
TOMM37
TOM complex
mitochondrion
Orf9b
precursor protein N-terminus binding
cell


Nsp7
Nsp5
Orf9c
Nsp8
Nsp4
TIMM29
NDUFAF7
ECSIT
ACAD9
NDUFB9
NDUFA1
MRPL3
MRPS22
MRPS16
Mt ribosomal proteins
active
MRPS5
MRPS2
MRPS25
MRPS27
Nsp8-affected Mt ribosomal proteins
TUFM
TSFM
GFM1
C12orf65
TWNK
Mt translation
POLG
MTPAP
POLG2
MTFMT
Mt replication
TFAM
active
TIMM9
TIMM10
TIM9-TIM10 complex
TIMM29
TIMM22
SDHC
TIM22 complex
TIMM23
TIMM17A
TIMM17B
TIM23 complex
TIMM9
TOMM22
TOMM70
TOMM37
TOM complex
Orf9b
ACAD9
NUBPL
COA1
COA7
DMAC1
DMAC2
ECSIT
FOXRED1
NDUFAF1
NDUFAF2
NDUFAF3
NDUFAF4
NDUFAF5
NDUFAF6
NDUFAF7
NDUFAF8
C1 assembly factors
CMC1
COA1
COA3
COA4
COA5
COA6
COA7
COA8
COX10
COX11
COX14
COX15
COX16
COX17
COX18
COX19
COX20
CHCHD7
HIGD1A
HIGD2A
c4 assembly
BCS1L
COX7A2
HCCS
c3 assembly factors
LYRM7
NUBPL
FDXR
FDX2
Fe-S subunit assembly
SDHAF1
SDHAF2
C2 assembly factors
OXPHOS factors
mtDNA encoded OXPHOS units
O_sub_2_endsub_
mt DNA replication
ERCC6
ERCC8
POLG
Mt-DNA repair
DGUOK
TK2
SUCLA2
SUCLG1
RRM2B
Mt-dNTP pool
mt DNA damage
precursor protein N-terminus binding
NDUFA1
NDUFA2
NDUFA3
NDUFA4
NDUFA5
NDUFA6
NDUFA7
NDUFA8
NDUFA9
NDUFA10
NDUFA11
NDUFA12
NDUFA13
NDUFAB1
alpha supernumerary subunits
NDUFB1
NDUFB2
NDUFB3
NDUFB4
NDUFB5
NDUFB6
NDUFB7
NDUFB8
NDUFB9
NDUFB10
NDUFB11
beta supernumerary subunits
NDUFS4
NDUFS5
NDUFS6
Fe-S supernumerary subunits
NDUFV1
NDUFV2
NDUFV3
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND5
MT-ND6
MT-ND4L
NDUFS1
NDUFS2
NDUFS3
NDUFS7
NDUFS8
e-
Complex 1
paraquat
H+
paraquat dication
MT-CO1
MT-CO2
MT-CO3
COX4I1
COX4I2
COX5A
COX5B
COX6A1
COX6A2
COX6B1
COX6B2
COX6C
COX7A1
COX7A2
COX7B
COX7B2
COX7C
COX8A
COX8C
e-
complex 4
SIRT3
NADH
AARS2
KARS1
GARS1
RARS2
DARS2
Mt-tRNA synthetase
TRMT1
mt DNA
MT-TC
MT-RNR1
MT-TH
MT-TQ
MT-TL1
MT-TS2
MT-TP
MT-TW
MT-TS1
MT-TF
MT-TV
MT-TK
MT-TE
MT-TN
MT-TI
MT-TR
MT-TL2
MT-TT
MT tRNAs
TFB1M
TFB2M
POLRMT
MT transcription
damaged mt DNA
Nsp6
ATP5MG
SDHA
SDHB
FAD
Fe-S
flavoprotein
Fe-S protein
SDHC
SDHD
complex 2
TCA
SOD2
NAD+
paraquat
ATP5F1A
ATP5F1B
ATP5F1C
ATP5F1D
ATP5F1E
ATP5F1EP2
F(1)
ATP5MC1
ATP5MC2
ATP5MC3
ATP5MG
ATP5PB
ATP5PD
ATP5ME
ATP5MF
MT-ATP8
ATP5PO
ATP5PF
F(0)
ATP5MD
ATP5MPL
MT-ATP6
ATP5IF1
ATP Synthase
mitochondrion
SDHA
SDHB
FADH2
Fe-S
flavoprotein
Fe-S protein
SDHC
SDHD
complex 2
superoxide
mt mRNA
UQCRC1
UQCRC2
UQCR10
UQCR11
UQCRB
UQCRQ
BCS1L
UQCRH
e-
UQCRFS1P1
UQCRFS1
Fe-S
Fe-S subunit
MT-CYB
heme b
MT-CYB subunit
complex 3
Cyt C
QH_sub_2_endsub
H+
SOD1
MT-ND1
MT-ND2
MT-ND3
MT-ND4
MT-ND4L
MT-ND5
MT-ND6
C1
MT-CO1
MT-CO2
MT-CO3
C4
MT-CYB
C3
MT-ATP8
MT-ATP6
C5
mtDNA encoded OXPHOS units
reduced
Cyt C
O_super_2-_endsuper_
Q
H_sub_2_endsub_O_sub_2_endsub_
ATP
H+
PRDX1
PRDX2
PRDX5
PRDX3
PRDX6
PRDX
TXNRD2
NADPH
GPX1
GPX4
Fe3+
Fe2+
TXN2
GSR
glutathione disulfide
hydroxide
CAT
NADP(+)
glutathione
HO
ROS
H_sub_2_endsub_O